AI 101 for Mechanical Engineers · Module 2

When ML is worth considering

Lesson 2.3

What you will learn

> Explain when ML may add value to an engineering project, use a simple baseline, check input and data suitability, and interpret a hybrid residual model.

Lesson overview

Module 2 · Lesson 3

When ML is worth considering

AI 101 for Mechanical Engineers

Start with the simplest method that meets the task.

If a trusted equation is accurate and fast enough, ML may add work without enough benefit.

1 When might ML help?

- Useful patterns are difficult to describe directly.
- Suitable observations represent the intended use.
- An expensive simulator needs a faster approximate model.

These are reasons to evaluate ML, not guarantees of success.

2 Four checks before proposing ML

1. **Purpose**
   What decision and output are needed?
2. **Inputs**
   Will they be available at decision time?
3. **Baseline**
   What simple method will you compare against?
4. **Coverage**
   Do the data represent the intended operating conditions?

Also state the project scope and how success will be judged.

3 Hybrid pump example

$$P_{\text{in}} = \frac{\Delta p \times Q}{\eta}$$

$\Delta p = 100{,}000$ Pa
$Q = 0.02$ m^3/s
$\eta = 0.8$ (dimensionless)

Hydraulic power
2,000 W

Predicted input power
2,500 W

Measured input power
2,800 W

Residual
$r = y - \hat{y} = 2{,}800\ \text{W} - 2{,}500\ \text{W} = +300\ \text{W}$

y: measured input power · $\hat{y}$: predicted input power · r: residual (W)

One residual does not show that the correction works elsewhere.

Check the benefit before adding complexity	Compare with the baseline.	Check physical assumptions against calibrated measurements.	**Add complexity only when evidence shows a useful improvement.**

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Start with the simplest adequate baseline

- A baseline is a simple comparison method [16]
- Begin with the simplest approach adequate for the decision
- Trusted equation meets required accuracy and speed?
- Then extra ML can add maintenance burden without sufficient benefit

When ML may be worth evaluating

- Relevant patterns cannot be specified directly
- Observations are suitable for the intended use
- A costly simulator needs an approximate response model [20]
- These justify evaluation; they do not guarantee success

Four checks before proposing ML

1. Purpose: state the engineering decision and required output
2. Inputs: available when the decision is made
3. Baseline: retain a simple comparison method
4. Coverage: measurements represent the intended operating range

Data must cover the conditions at use

- Also state the success criterion and project scope
- One machine or condition may not represent another [21]
- Consider loads, operating modes, shifts and seasonal environments
- State the operating domain; gather evidence for conditions that matter

The hybrid keeps the physical baseline

$$P_{in} = \Delta p \times Q / \eta$$

$$P_{in} = 100{,}000\ \text{Pa} \times 0.02\ \text{m}^3/\text{s} / 0.8 = 2{,}500\ \text{W}$$

$$r = y - \hat{y} = 2{,}800\ \text{W} - 2{,}500\ \text{W} = +300\ \text{W}$$

- Hydraulic power 2,000 W; predicted input power 2,500 W
- A hybrid can learn a correction from residuals [14]
- One residual does not show the correction applies elsewhere

Validate the proposed benefit

Compare with baseline ▶ **Check physical assumptions** ▶ **Check operating coverage** ▶ **Add justified complexity**

- Check assumptions against calibrated measurements
- Add complexity only when evidence shows useful accuracy, speed or outcome gains

CHECK YOUR UNDERSTANDING

What must be stated before proposing ML for an engineering project?

Answer before moving on.

CHECK YOUR UNDERSTANDING

What must be stated before proposing ML for an engineering project?

> The decision, inputs available at use time, required output, success criterion, a comparison baseline and whether representative data are available.

Key takeaways

- Use a simple adequate baseline before adding ML
- Check purpose, available inputs, baseline and operating coverage
- Hybrid models keep physical structure and learn from discrepancies
- Validate both data coverage and physical assumptions

References

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Numbers match the course reference list. The written lesson is the authoritative text.